

Electric Circuits II Project

Report

Prepared by

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TutorialNumber:

T11

Theoretical background:

1. Filter Overview

A filter is a frequency-selective circuit designed to pass signals within a desired frequency band while rejecting or attenuating signals outside that band. The chosen circuit is an **Active Inverting Band Pass Filter**.

As an active filter, it utilizes an operational amplifier (the **LM324**) alongside passive resistors and capacitors to provide both frequency selectivity and signal gain. This specific configuration combines the characteristics of:

- **A High-Pass Section:** Formed by the series combination of R_1 and C_1 at the input stage.
- **A Low-Pass Section:** Formed by the parallel combination of R_2 and C_2 in the feedback loop.

The resulting band-pass characteristic passes frequencies between a lower cutoff frequency (f_L) and a higher cutoff frequency (f_H).

2. Design Equations

The performance of the filter is determined by the values of the passive components according to the following mathematical relationships provided in the project description:

A. Cutoff Frequencies

The half-power frequencies, or cutoff frequencies, define the boundaries of the passband where the output power drops to half its maximum value (or -3 dB):

- **Low Cutoff Frequency (f_L):** Sets the start of the passband.

$$f_L = \frac{1}{2\pi R_1 C_1}$$

- **High Cutoff Frequency (f_H):** Sets the end of the passband.

$$f_H = \frac{1}{2\pi R_2 C_2}$$

For a proper band-pass response, the components must be chosen such that $f_L < f_H$.

B. Filter Gain

The gain of the filter represents the ratio of the output voltage to the input voltage:

$$Gain = \frac{V_{out}}{V_{in}}$$

In the mid-band (passband) region, the maximum gain for this inverting configuration is approximately determined by the ratio of the feedback resistance to the input resistance:

$$|A_v| \approx \frac{R_2}{R_1}$$

According to the design guidelines, the gain should be maintained between **1 and 10**.

C. Center Frequency (f_o)

The center frequency represents the geometric mean of the two cutoff frequencies, where the filter typically reaches its peak gain:

$$f_o = \sqrt{f_L \cdot f_H}$$

Filter design:

1. Component Values

These values were used in both the PSpice simulation and for the hardware implementation on the breadboard

Component	Label	Value
Resistor 1	R_1	1 k Ω
Resistor 2	R_2	2.4 k Ω
Capacitor 1	C_1	3.3 nF
Capacitor 2	C_2	1 nF
Op-Amp	U_1	LM324

2. Design Parameters

The following parameters were calculated using the theoretical formulas for the **Inverting Band Pass Filter** (Circuit 1):

- **Lower Cutoff Frequency (f_L):**

$$f_L = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi(1000)(3.3 \times 10^{-9})} \approx \mathbf{48.23 \text{ kHz}}$$

- **Higher Cutoff Frequency (f_H):**

$$f_H = \frac{1}{2\pi R_2 C_2} = \frac{1}{2\pi(2400)(1 \times 10^{-9})} \approx \mathbf{66.31 \text{ kHz}}$$

- **Passband Gain:**

$$|A_v| = \frac{R_2}{R_1} = \frac{2400}{1000} = \mathbf{2.4}$$

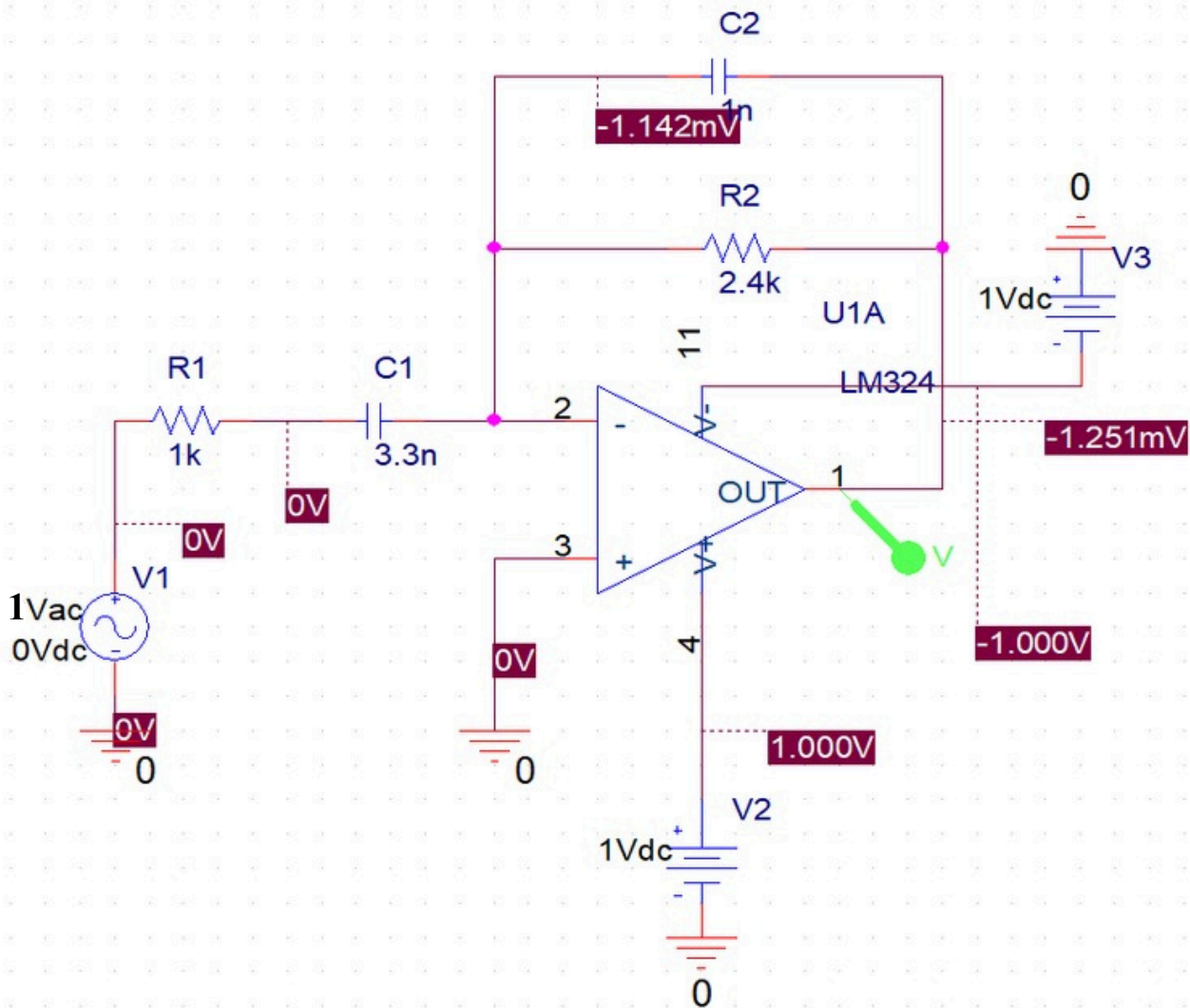
- **Center Frequency (f_o):**

$$f_o = \sqrt{f_L \cdot f_H} \approx \mathbf{56.55 \text{ kHz}}$$

Circuit simulation:

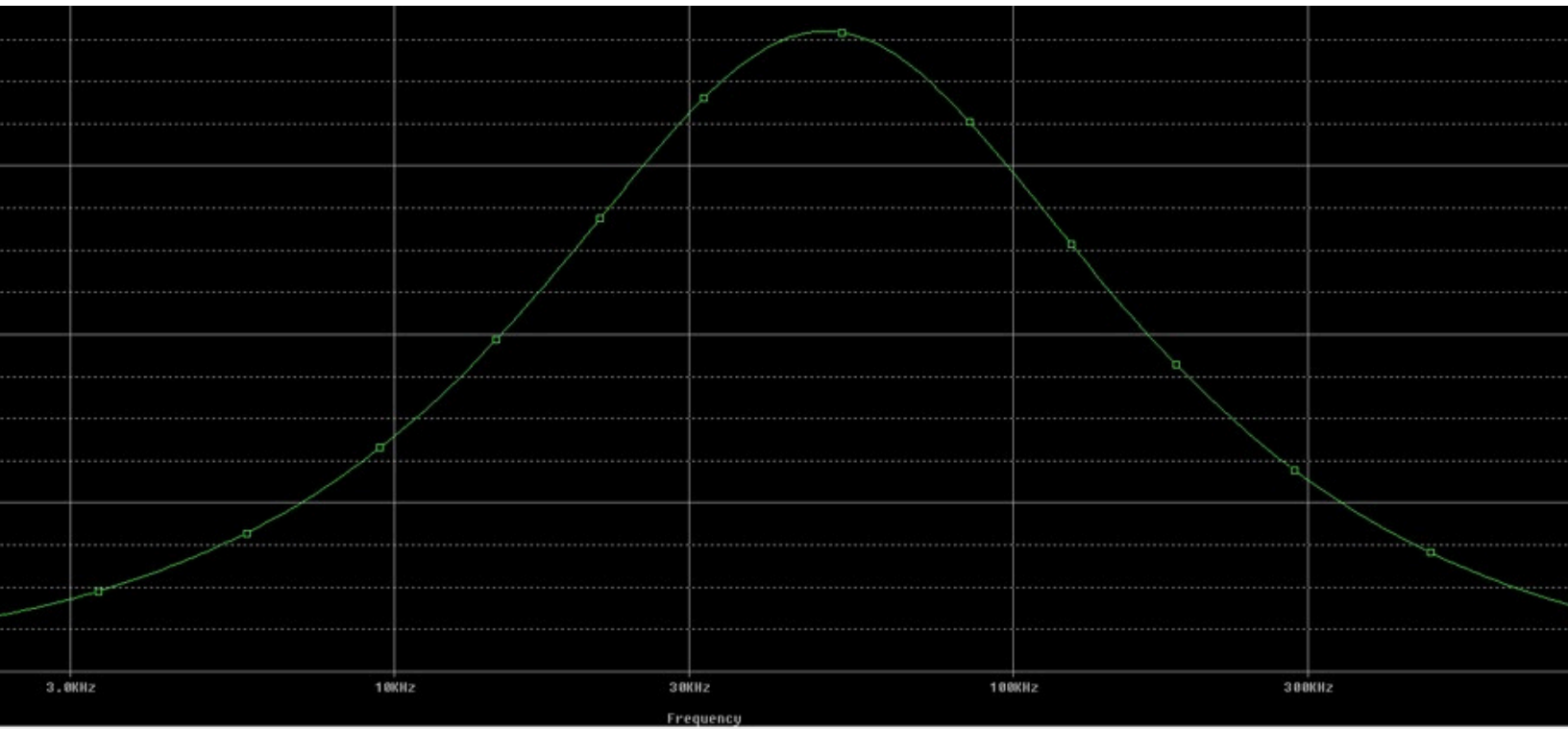
Includes the detailed steps for the circuit construction on the software, i.e. screenshots for the schematic diagram and simulation profiles.

NB: You can refer to any lab manual which included simulations on PSpice for aid.

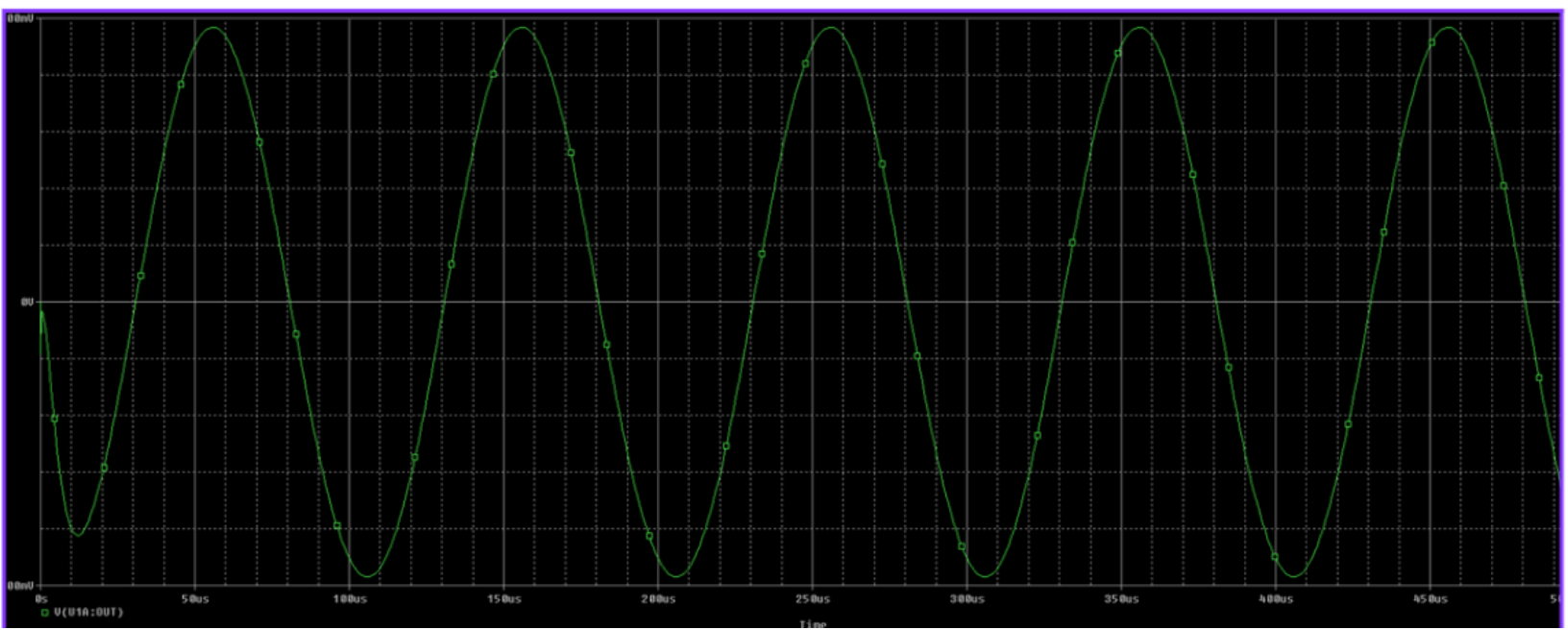


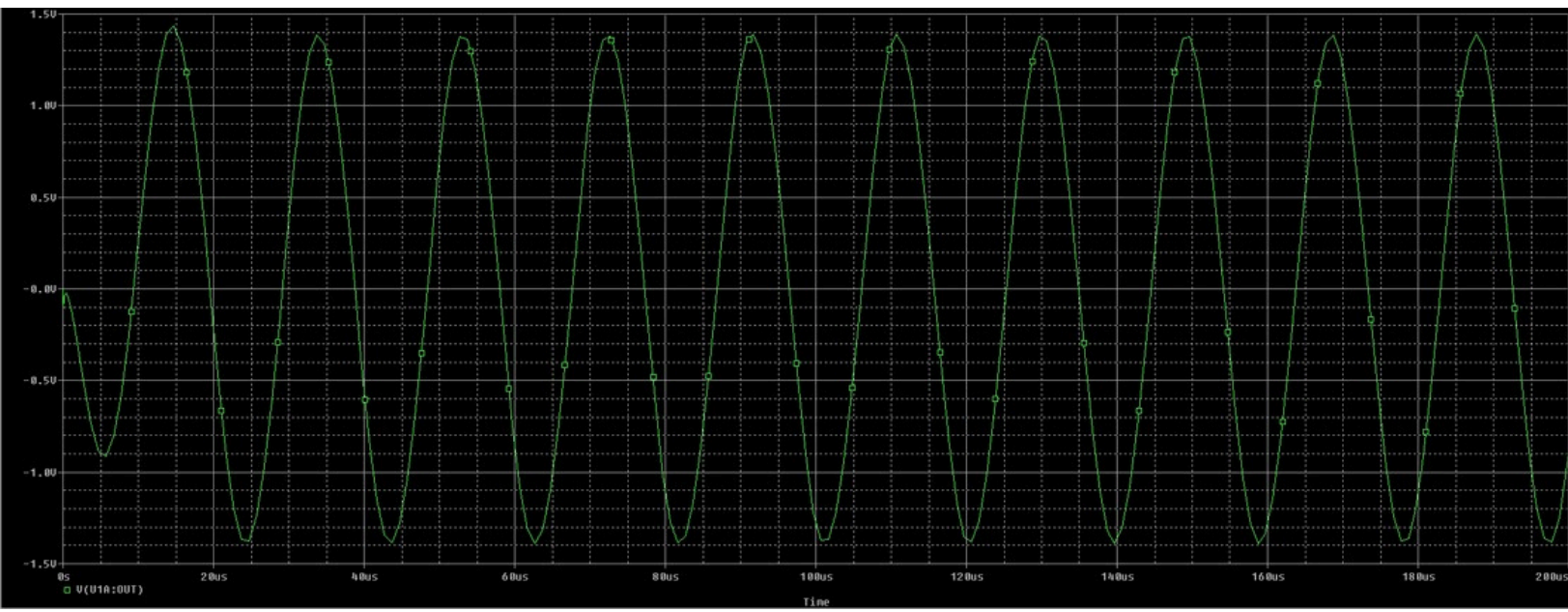
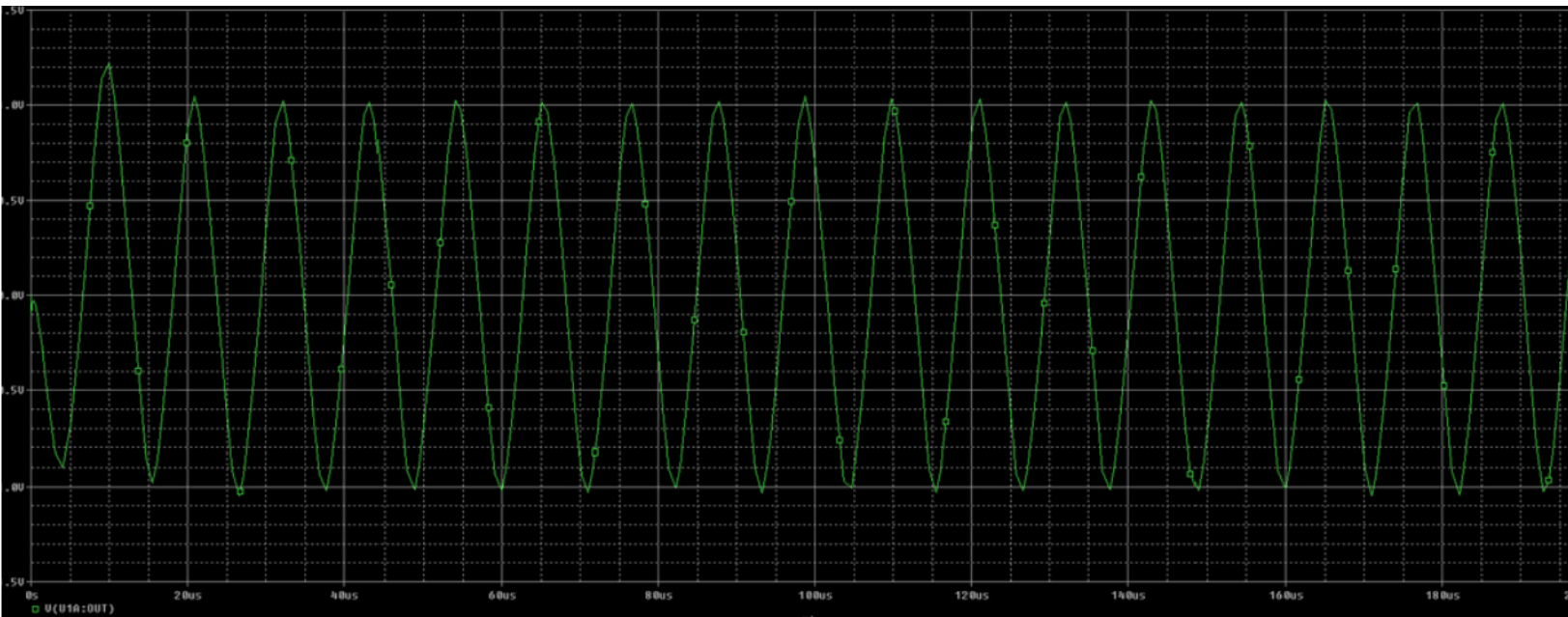
Simulation results:

i. AC sweep simulation:

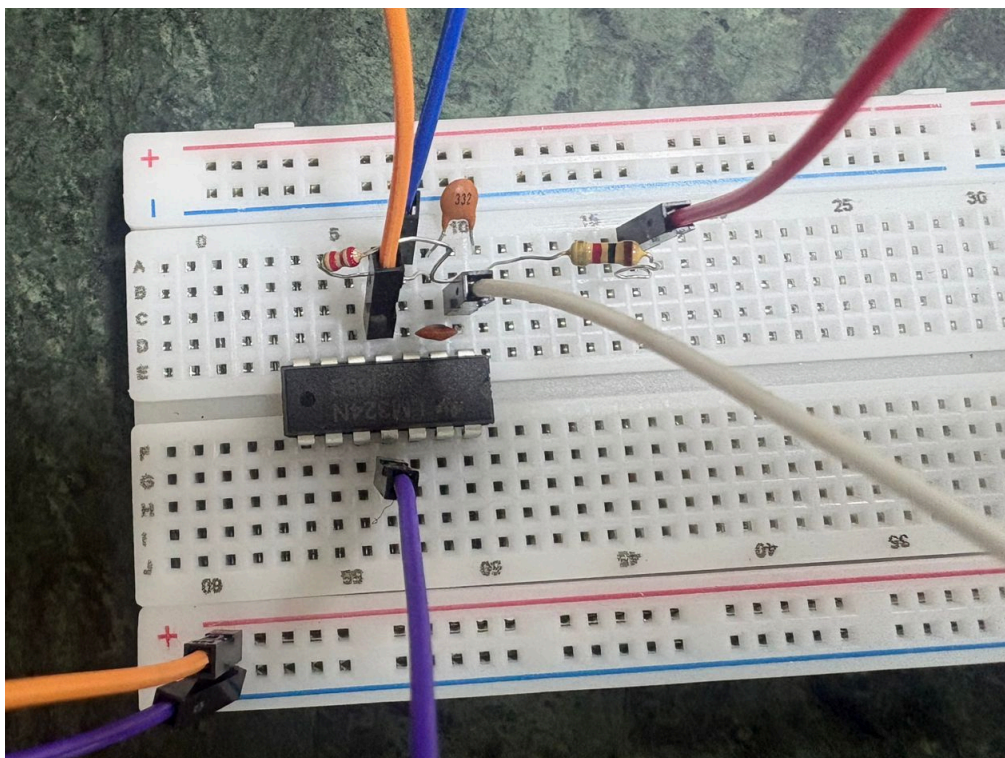
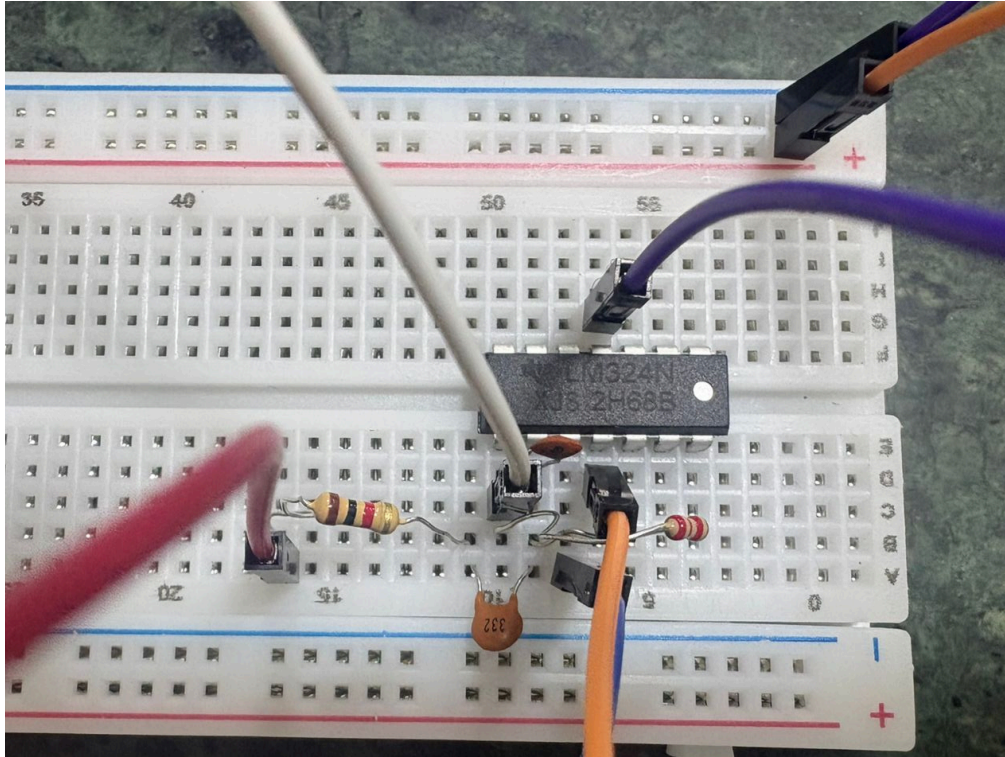


ii. Time domain simulation:

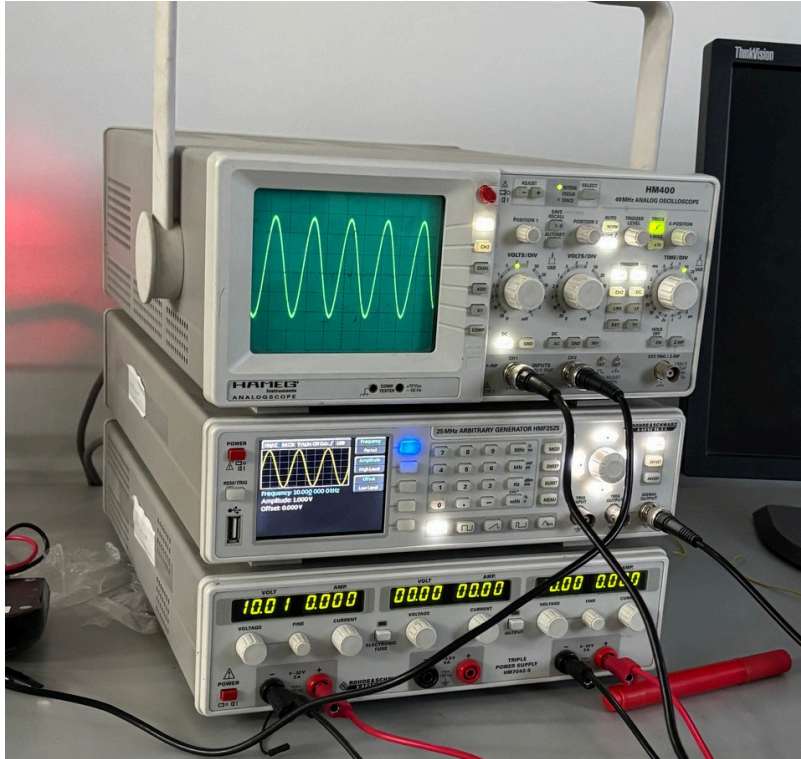




Hardware implementation:



Hardware results:



i. $f < f_0$: **10kHz**

ii. $f = f_0$: **56.6kHz**

iii. $f > f_0$: **90kHz**